

Mathematical Foundation for Data Science (SC613)

Course Placement: It is a core course for M. Sc. (Data Science) 1st semester.

Course format: Three hours' lecture and two-hours tutorial every week.

Course Content:

S. No.	Unit Title	Topics	Lectures
1	Probability	(a) Basic of Probability: Axioms of Probability, Independence, Conditional Probability, Bayes theorem.	6
		(b) Random variables: Random Variables, Discrete and Continuous Random variables, Expectation, Mean and Variance of Random variables, and functions of one random variable.	4
		(c) Probability Density/Distribution, Cumulative Density/Distribution functions: Definition and some examples: Uniform and Gaussian.	5
2	Linear Algebra	(a) Vectors: Vector Spaces, Examples, Operations on Vectors: Addition, Scaling, Linear Combination, Span, Subspaces, Basis, Linear Dependence and Independence. Span, Data as Vectors, Euclidean Space R^n , Inner product Space, Orthogonality, Norms in Euclidean Space R^n , Distances.	5
		(b) Matrices:- Matrices as Data, Rank of Matrix, Matrix operations, Addition, Multiplication, Inversion.	5
		(c) Matrices: Linear transformations their representation as matrices, Change of Basis - Similarity transformations	5
		(d). Eigenvalues and Eigenvectors. A special case of Similarity transformation, Characteristic polynomial for computing eigenvalues and eigenvectors. Matrix Diagonalization, Singular value decomposition, Principal Component Analysis.	5
3.	Calculus	Functions, function of several variables, continuous functions, convex functions, differentiable functions, gradient of a function, directional derivatives. Minima of a convex function, gradient descent method.	5

Suggested books:

- Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong. *Mathematics for machine learning*. Cambridge University Press, 2020

- A First Course in Probability, Sheldon Ross, Pearson, 10th ed., 2023
- Modern Statistics: Intuition, Math, Python, R, Mike X Cohen, Sincxpress, 2023.
- David C. Lay, Steven R. Lay and Judi McDonald, Linear Algebra and its Applications, Pearson, 5th ed., 2015.
- Linear Algebra for Data Science, Moshe Haviv, World Scientific, 2023.
- Linear Algebra for Everyone, Gilbert Strang, Wellesley-Cambridge Press, 2020.

Grading method: Evaluation through multiple class activities. The tentative scheme is

Assignments (30 %), multiple class quizzes (50 %), and End-Term Examination (20%)
(Subject to change as per the response of the class)

Course Outcomes: After completion of this course, students will

- develop the depth understanding of the fundamental mathematical concepts required for data science.
- develop the depth understanding of the fundamental statistical concepts required for data science.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X			X	X							